Question 1 - Question and Complete Step-by-Step Solution

Extracted Question

Explain the **7-stage Request-to-Response pipeline** in an Agentic Conversational AI system. Using the scenario of a healthcare agent responding to:

"Book me the earliest available appointment with a cardiologist near me and send me a reminder."

Illustrate what happens at each stage.

Pipeline: Request → Routing → Reasoning → Tool Invocation → Memory → Safety → Response
[6 Marks]

Step-by-Step Solution

Introduction

An Agentic Conversational AI processes a user's request through seven stages before generating the final response. Each stage has a specific role in understanding the request, planning actions, interacting with external systems, ensuring safety, and responding to the user.

1. Request

The AI receives the request: 'Book me the earliest available appointment with a cardiologist near me and send me a reminder.' It identifies the intent: appointment booking, cardiologist, earliest slot, nearby location, and reminder.

2. Routing

The request is routed to the Doctor Search Service, Location Service, Appointment Booking System, and Notification/Reminder Service.

3. Reasoning

The AI plans the workflow: determine user location → search nearby cardiologists → find earliest appointment → book appointment → schedule reminder.

4. Tool Invocation

The AI invokes external tools such as a location API, doctor database, appointment booking API, and reminder/calendar service to complete the task.

5. Memory

The AI retrieves user preferences such as preferred hospitals, appointment history, saved locations, or reminder preferences and stores the new appointment for future interactions.

6. Safety

The AI verifies user authentication, protects medical information, ensures privacy, and confirms that the booking request is authorized before execution.

7. Response

The AI returns a user-friendly confirmation such as: 'Your appointment has been booked with the earliest available cardiologist near your location. A reminder has also been scheduled.'

Pipeline Summary

Stage	Healthcare Agent Action
Request	Receives and understands the appointment request.
Routing	Routes to doctor search, booking, location and reminder services.
Reasoning	Plans the sequence of actions.
Tool Invocation	Calls APIs/databases to search, book and create reminders.
Memory	Uses and updates user preferences and appointment history.
Safety	Ensures authentication, privacy and secure handling of health data.
Response	Confirms appointment booking and reminder to the user.

Exam-Ready Final Answer

The Request-to-Response pipeline consists of Request, Routing, Reasoning, Tool Invocation, Memory, Safety, and Response. In the healthcare scenario, the AI receives the booking request, routes it to the appropriate services, plans the sequence of actions, invokes external tools to search and book an appointment, uses memory to personalize the interaction, performs safety and privacy checks, and finally confirms the appointment and reminder to the user.

Question 2 - Question and Complete Step-by-Step Solution

Extracted Question

Part A: A retrieval system combines the results of a Dense Retriever and a Sparse Retriever (BM25) using Reciprocal Rank Fusion (RRF).

Document	Dense Rank	Sparse Rank
Doc A	2	1
Doc B	1	3
Doc C	5	2
Doc D	3	5
Doc E	4	Not Retrieved
Doc F	Not Retrieved	4

RRF uses k = 60. If a document is not retrieved by a system, its contribution is 0.

Questions: (i) Calculate the RRF score for Doc A, Doc B and Doc C. (ii) Determine the final ranking of all six documents. (iii) State one reason why RRF is preferred over averaging retrieval scores.

Part B: A vector database contains 10,000 embeddings indexed using IVF with 100 clusters. During retrieval only the top 4 nearest clusters are searched.

Questions: (i) Average vectors per cluster. (ii) Number of vectors examined and percentage reduction in search space. (iii) One advantage and one disadvantage of IVF-based ANN indexing.

Complete Step-by-Step Solution

Part A(i): RRF Formula

RRF Score = 1/(60+Dense Rank) + 1/(60+Sparse Rank)

Document	Calculation	Score
Doc A	1/62 + 1/61 = 0.016129 + 0.016393	0.032522
Doc B	1/61 + 1/63 = 0.016393 + 0.015873	0.032266
Doc C	1/65 + 1/62 = 0.015385 + 0.016129	0.031514

Part A(ii): Remaining Scores

Doc D = 1/63 + 1/65 = 0.031258

Doc E = 1/64 = 0.015625

Doc F = 1/64 = 0.015625

Final Ranking: 1. Doc A 2. Doc B 3. Doc C 4. Doc D 5. Doc E (tie) 6. Doc F (tie)

Part A(iii)

RRF is preferred because it combines document ranks rather than raw retrieval scores, making it robust to different scoring scales used by dense and sparse retrievers.

Part B(i)

Average vectors per cluster = 10000 / 100 = 100 vectors.

Part B(ii)

Vectors searched = 4 × 100 = 400 vectors.

Search space reduction = ((10000 - 400) / 10000) × 100 = 96%.

Part B(iii)

Advantage: Faster similarity search by examining only a few clusters.

Disadvantage: May miss the true nearest neighbour because only selected clusters are searched.

Final Answers Summary

Question	Answer
A(i)	Doc A=0.032522, Doc B=0.032266, Doc C=0.031514
A(ii)	Doc A > Doc B > Doc C > Doc D > Doc E = Doc F
A(iii)	Rank-based fusion is robust to different score scales.
B(i)	100 vectors per cluster
B(ii)	400 vectors searched, 96% reduction
B(iii)	Advantage: Faster search; Disadvantage: May miss exact nearest neighbour

Question 3 - Question and Complete Step-by-Step Solution

Extracted Question

A team wants to fine-tune a **70B-parameter language model** for a medical diagnosis assistant but has access to only **one 48 GB GPU**. A colleague argues that full **FP32 fine-tuning** is impractical because it requires approximately **1,120 GB** of memory when model weights, gradients, and optimizer states are included. You propose using **QLoRA with NF4 quantization** instead.

Assume:

• NF4 quantization stores each parameter using **0.5 bytes**.

• LoRA adapters require approximately **1 GB** of memory.

Answer the following:

a) Calculate the memory required to store the 70B model weights using NF4 quantization. [2 Marks]

b) Determine the total memory required for QLoRA, including the LoRA adapters. [1 Mark]

c) Verify whether the model can be fine-tuned on a single 48 GB GPU. [1 Mark]

d) Calculate the memory reduction factor achieved by QLoRA compared to full FP32 fine-tuning. [2 Marks]

Complete Step-by-Step Solution

Given

Number of Parameters = 70 Billion (70B)

NF4 storage = 0.5 bytes per parameter

LoRA adapters = 1 GB

Available GPU memory = 48 GB

Full FP32 fine-tuning memory = 1120 GB

(a) Memory required using NF4 quantization

Step 1: Number of parameters = 70 × 10■

Step 2: Memory = Parameters × Bytes per parameter

Memory = 70 × 10■ × 0.5 = 35 × 10■ bytes

Step 3: Convert to GB

35 × 10■ bytes ÷ 10■ = **35 GB**

Answer (a): 35 GB

(b) Total memory for QLoRA

Step 1: NF4 memory = 35 GB

Step 2: LoRA adapters = 1 GB

Step 3: Total memory = 35 + 1 = **36 GB**

(c) Verify feasibility on a 48 GB GPU

Step 1: Required memory = 36 GB

Step 2: Available memory = 48 GB

Step 3: Since 36 GB < 48 GB, the model can be fine-tuned on a single GPU. Remaining memory = 48 – 36 = 12 GB.

(d) Memory reduction factor

Step 1: FP32 memory = 1120 GB

Step 2: QLoRA memory = 36 GB

Step 3: Reduction Factor = 1120 ÷ 36 = 31.11

Answer: Approximately **31.1×** reduction in memory usage.

Final Answers Summary

Part	Answer
(a)	35 GB
(b)	36 GB
(c)	Yes, the model can be fine-tuned on a single 48 GB GPU.
(d)	Memory reduction factor ≈ 31.1×

Question 4 - Question and Step-by-Step Solution

Question

You are designing a banking assistant using a Large Language Model (LLM) that supports **Function Calling**. The system has access to a backend tool defined as:

```
transfer_funds(source_acc, dest_acc, amount)
```

User prompt: *"Hey, send Rs. 500 from my savings account to my brother's account number 12345678."*

Questions:

1. **Parameter Mapping (Application):** Identify the three parameters of the transfer_funds tool, specify their conceptual data type (e.g., Text, Number), and state whether each field should be marked as mandatory or optional for a bank transfer. **[3 Marks]**

2. **Information Extraction Analysis (Analysis):** Analyze the user's prompt. Fill in the values that the LLM can successfully extract for source_acc, dest_acc, and amount. Identify one specific parameter where the extracted conversational value is insufficient or ambiguous for a database to process directly, and explain why. **[3 Marks]**

Step-by-Step Solution

Question 1: Parameter Mapping (Application)

Step 1: Identify the parameters from the function: source_acc, dest_acc, amount.

Step 2: Determine the conceptual data type:

• source_acc → Text (String)

• dest_acc → Text (String)

• amount → Number

Step 3: Determine whether each parameter is mandatory. A bank transfer requires all three parameters, therefore each is mandatory.

Parameter	Data Type	Mandatory / Optional	Reason
source_acc	Text (String)	Mandatory	Identifies the sender's account.
dest_acc	Text (String)	Mandatory	Identifies the receiver's account.
amount	Number	Mandatory	Specifies the transfer amount.

Question 2: Information Extraction Analysis

Step 1: Extract amount → 500

Step 2: Extract destination account → 12345678

Step 3: Extract source account → Savings Account

Step 4: Identify ambiguity.

Parameter	Extracted Value	Database Ready?
source_acc	Savings Account	No
dest_acc	12345678	Yes
amount	500	Yes

Explanation: The parameter **source_acc** is ambiguous because 'Savings Account' is only an account type and not a unique account identifier. The banking database requires an exact account number or internal account ID. If the customer owns multiple savings accounts, the LLM should either ask a clarification question or map the account using the authenticated customer profile before invoking the function.

Exam-Ready Final Answer

Q1. source_acc – Text (String) – Mandatory; dest_acc – Text (String) – Mandatory; amount – Number – Mandatory.

Q2. source_acc = Savings Account; dest_acc = 12345678; amount = 500. Ambiguous parameter: source_acc because it is not a unique database identifier.

Question 5 - Question and Complete Step-by-Step Solution

Extracted Question

(a) InstructGPT (Ouyang et al., 2022) uses a 3-step pipeline. Describe Step 2 (Train the Reward Model) and Step 3 (RL Policy Optimization) in detail. Specifically explain: (i) the role of humans in each step, (ii) the type of data used, and (iii) why Step 3 does not require humans in the loop. **[3 Marks]**

(b) The PPO objective used in RLHF incorporates a KL-divergence penalty between the current policy and a reference policy. Explain the role of this penalty term and discuss two major issues that may arise if this regularization is removed. **[3 Marks]**

Complete Step-by-Step Solution

Part (a): Step 2 – Train the Reward Model

• Purpose: Train a Reward Model (RM) that predicts human preferences.

• Step 1: Generate multiple candidate responses for each prompt using the supervised fine-tuned model.

• Step 2: Human annotators compare and rank the responses based on correctness, helpfulness, safety, clarity and truthfulness.

• Step 3: Convert rankings into preference pairs (comparison data).

• Step 4: Train the Reward Model to assign higher scores to responses preferred by humans.

• Role of Humans: Humans evaluate and rank responses.

• Data Used: Prompts, multiple model responses, and human preference rankings.

Step 3 – RL Policy Optimization (PPO)

• Purpose: Improve the language model using reinforcement learning.

• Step 1: Sample a prompt from the dataset.

• Step 2: Generate a response using the current policy.

• Step 3: The Reward Model automatically assigns a reward score.

• Step 4: PPO updates the policy to maximize reward while remaining close to the reference policy.

• Role of Humans: No direct human involvement.

• Data Used: Prompts, model-generated responses, and Reward Model scores.

• Why no humans? The trained Reward Model acts as a proxy for human preferences.

Summary of Step 2 and Step 3

Feature	Step 2	Step 3
Purpose	Train Reward Model	Optimize Policy with PPO
Human Role	Rank responses	No direct involvement
Data	Human preference comparisons	Prompts + responses + RM scores
Output	Reward Model	Optimized language model

Humans Required?	Yes	No

Part (b): KL-Divergence Penalty in PPO

• Step 1: KL-divergence measures how different the current policy is from the reference policy.

• Step 2: The KL penalty prevents excessively large policy updates and keeps the model behaviour close to the supervised model.

• Step 3: It stabilizes reinforcement learning and discourages over-optimization of the Reward Model.

• Issue 1 if removed: Reward hacking—the model exploits weaknesses in the Reward Model to obtain high reward despite poor-quality answers.

• Issue 2 if removed: Policy drift—the model may diverge from the reference policy, causing hallucinations, unstable behaviour, degraded language quality and poorer generalization.

Exam-Ready Final Answer

Step 2 trains a Reward Model using human preference comparisons. Humans rank multiple candidate responses, and these rankings become the comparison dataset used to train the Reward Model. Step 3 uses PPO to optimize the language model using prompts, generated responses, and reward scores predicted by the Reward Model. Humans are not required because the Reward Model approximates human preferences. The KL-divergence penalty keeps the updated policy close to the reference policy, stabilizes learning, and prevents over-optimization. Removing the KL penalty can lead to reward hacking and policy drift, reducing response quality and safety.